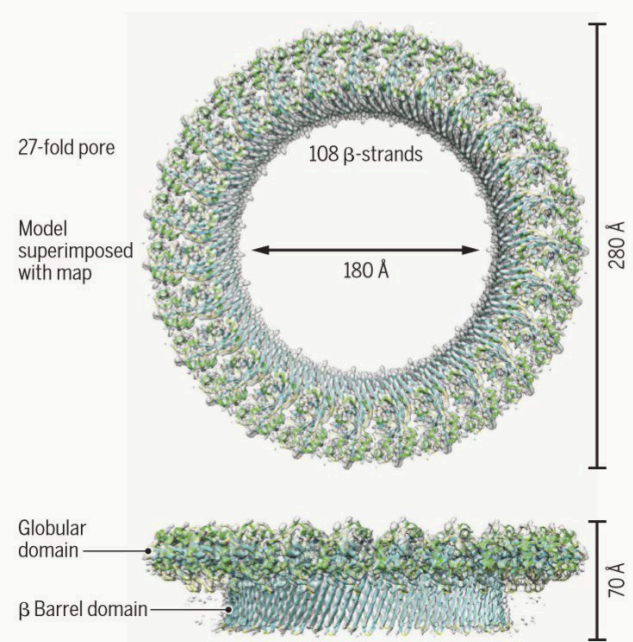


Talking Between Cells

CD molecules, MHC, cytokines, pattern-recognition receptors and the inflammasome, read as one conversation.

Nothing commits on one signal.

This is the observation that ties Lecture 3 to Lecture 4. A T cell and a macrophage face the same risk: acting on a single, possibly mistaken, signal. Both solve it the same way.



The gasdermin D pore. A ring of 108 β -strands, 180 Å across. It is the door IL-1 β leaves by, and it opens only after two signals. See page 4.

	SIGNAL 1	SIGNAL 2	SIGNAL 3
	Is it there?	Is it dangerous?	What kind of response?
In a T cell	TCR sees peptide on MHC	CD28 meets B7 on an activated APC	Cytokines steer it to Th1, Th2, Th17 or Treg
In a macrophage	A PRR sees a PAMP, NF- κ B writes pro-IL-1 β	ATP, a crystal or a toxin assembles NLRP3	IL-1 β and IL-18 recruit and instruct the neighbours
One alone	T cell: anergy, permanently switched off. Macrophage: primed, releases nothing.		Not needed to commit, only to steer

p. 2

The three signals: how a T cell is switched on

p. 3

MHC: the display case, hot dog and pita

p. 4

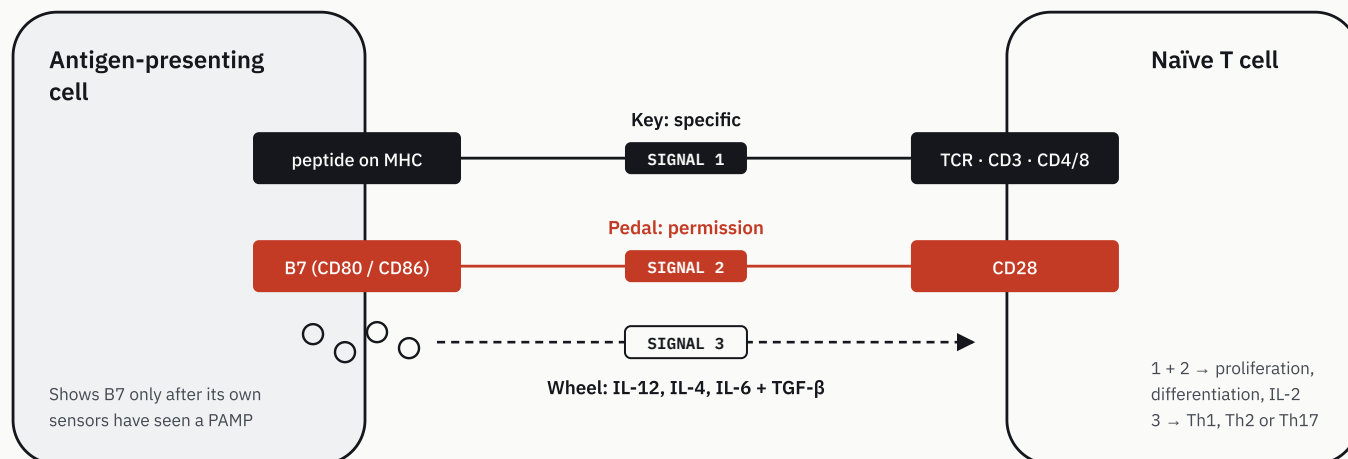
The inflammasome: two signals, again

p. 5

Tell them apart, and test yourself

A T cell needs a key, a pedal and a wheel.

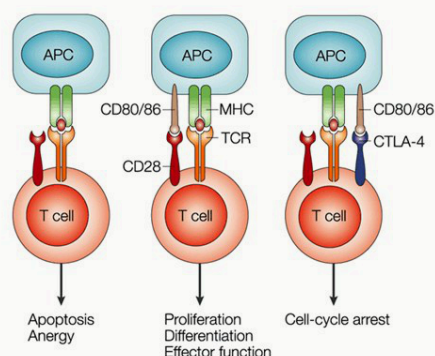
Meeting its antigen does not decide what a T cell does. A second molecule on the other cell decides whether it responds, does nothing, or is disabled for good.



Anergy

Signal 1 without signal 2. The cell survives but is **functionally inactivated**, unresponsive to that antigen even if it is later presented with full costimulation.

Resting tissue cells show self-peptides all day and carry no B7. A self-reactive T cell that escaped the thymus is silenced by that meeting, not activated by it. The two-signal rule is **peripheral tolerance**.



Nature Reviews | Immunology

Same APC, three outcomes. TCR only: anergy. Plus CD28: proliferation. Plus CTLA-4: cell-cycle arrest.

CD28 and CTLA-4: same ligand, opposite job

	CD28 the accelerator	CTLA-4 (CD152) the brake
Present	Constitutive, already on the resting T cell	Induced, only after TCR stimulation
Binds B7	Lower affinity	Much higher affinity, outcompetes CD28
Effect	Proliferation, differentiation, IL-2	Cell-cycle arrest, reduced IL-2
If lost	No costimulation	Mice die of lymphoproliferation within 3-4 weeks
Logic	Switching a T cell on also switches on the molecule that will switch it off. The response is built to end.	

The TCR signal runs Lck → ZAP-70 → PLC γ 1 → calcineurin → NFAT → IL-2 gene. Cyclosporin and tacrolimus block calcineurin, which is how graft rejection is prevented.

Releasing the brake on purpose

Ipilimumab, an anti-CTLA-4 antibody, lets T cells attack melanoma; James P. Allison shared the 2018 Nobel Prize for the idea. Its side effects (colitis, thyroiditis, hepatitis, dermatitis) are the proof: you have removed peripheral tolerance from the whole body, not only from the tumour.

Class I is a hot dog. Class II is a pita.

An MHC molecule is a display case. Ask where its peptide came from, and the class, the co-receptor and the T cell that reads it all follow.



MHC class I

HLA-A, -B, -C

8-10

amino acids.
Bun closed at both ends, so the sausage is cut to fit.

Cleft walls	$\alpha 1 + \alpha 2$, one chain
Peptide from	Inside the cell: cytosolic proteins cut by the proteasome, pumped into the ER by TAP
Found on	Every nucleated cell. The message: here is what I am making
Read by	CD8 cytotoxic T cells; CD8 grips $\alpha 3$



MHC class II

HLA-DR, -DQ, -DP

13-25

amino acids.
Open at both ends, so the filling hangs out.

Cleft walls	$\alpha 1 + \beta 1$, two chains
Peptide from	Outside: endocytosed proteins chewed by cathepsins in the acidified endosome
Found on	B cells, dendritic cells, macrophages. The message: here is what I found
Read by	CD4 helper T cells; CD4 grips $\beta 2$

Inside, *eight*. Outside, *four*.

Class I shows what is made inside the cell, to CD8. Class II shows what came from outside, to CD4. The shapes are not arbitrary: the proteasome supplies size-matched peptides, so a closed pocket works; cathepsins leave ragged lengths, so only an open groove can hold them.

The rule of eight

CD8 $\times$ class I = **8**

CD4 $\times$ class II = **8**

Two arithmetic facts rebuild the whole table.

HLA alleles you may be asked to test for

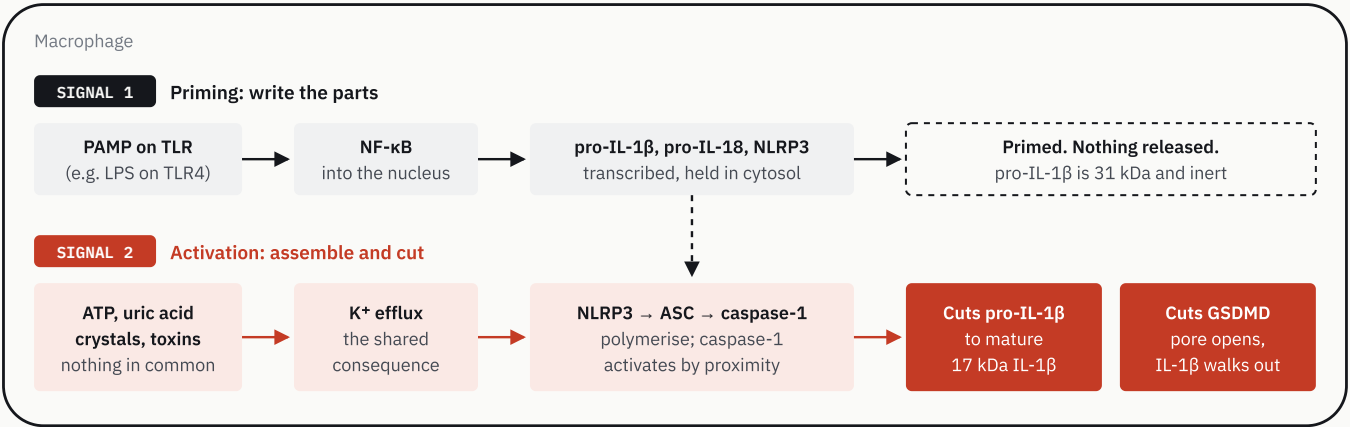
HLA-B27	Ankylosing spondylitis, reactive arthritis, anterior uveitis. A risk allele, not a diagnosis
HLA-DQ2 / DQ8	Coeliac disease. Their absence essentially rules it out
HLA-DR3 / DR4	Type 1 diabetes

HLA-B*57:01	Abacavir hypersensitivity. Testing before prescribing is mandatory in HIV care
HLA-B*15:02	Carbamazepine-induced Stevens-Johnson syndrome in Han Chinese
HLA-DR15	Multiple sclerosis

Exam trap. $\beta 2$ -microglobulin is not an MHC gene. It sits on chromosome 15, is invariant, and never causes transplant mismatch.

The inflammasome: two signals, again.

IL-1 β can cause fever and shock. It has no signal peptide and is made inactive, so a macrophage must hear two independent signals before it can release a single molecule.



Three parts, three jobs

Sensor

NLRP3

LRR NACHT PYD

Adaptor, a pure connector

ASC

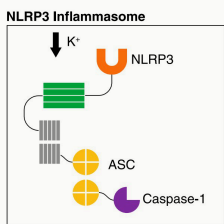
PYD CARD

Executioner

pro-caspase-1

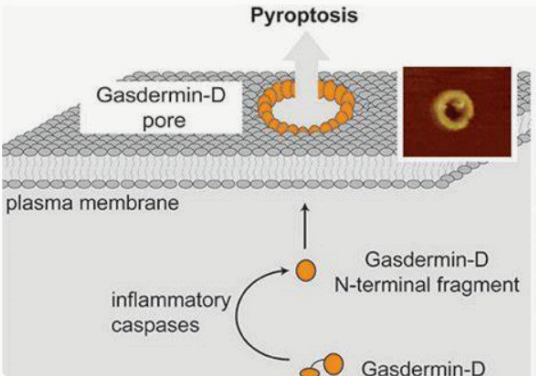
CARD p20 p10

NLRP3 does not bind its activators. It detects a state they all cause, mostly potassium loss. It is a sensor of cellular distress, which is why it shows up in gout, atherosclerosis and type 2 diabetes.



Seen down a microscope as one large perinuclear **ASC speck** per cell, the readout that it has fired.

How IL-1 β actually leaves: gasdermin D



- 1 Caspase-1, 4, 5 or 11 cuts GSDMD, freeing its N-terminal fragment. Apoptotic caspases cannot.
- 2 The fragments form a 27-fold ring pore in the plasma membrane, about 180 Å wide.
- 3 Mature IL-1 β and IL-18 pass out through it.
- 4 Enough pores and the cell bursts: **pyroptosis**, loud where apoptosis is silent.

The parallel to say out loud

	T cell (Part 2)	Macrophage (Part 7)
SIGNAL 1	TCR sees peptide on MHC. Specific	PRR sees a PAMP. Specific
SIGNAL 2	CD28 sees B7. Permission	ATP, crystal or toxin. Permission
One alone	Energy	A primed cell that releases nothing

Tell them apart, then test yourself.

Most lost marks come from look-alikes. Each pair below is separated by one fact.

CD28 versus CTLA-4	Same ligand, B7. CD28 is constitutive and stimulatory; CTLA-4 is induced, higher-affinity and inhibitory	MHC I versus MHC II	Inside, eight; outside, four. $CD8 \times I = CD4 \times II = 8$
β2-microglobulin versus class II β chain	β 2m is on chromosome 15 and invariant. The class II β chain is an MHC gene and polymorphic	γc versus JAK3	The receptor chain versus the kinase it carries. Either loss gives SCID, but γ c is X-linked and JAK3 autosomal recessive
PAMP versus DAMP	Microbial molecule versus host molecule in the wrong place. Same receptors, same cytokines	MyD88 versus TRIF	MyD88 serves every TLR except TLR3. TLR3 uses TRIF
Caspase-1 versus caspase-3	Inflammatory, cuts IL-1 β and GSDMD, versus apoptotic. Same family, opposite meaning	Apoptosis versus pyroptosis	Membrane intact and silent, versus membrane ruptured by GSDMD pores and maximally inflammatory

Self-test

Answer out loud before reading on. An answer you can recognise but not produce is not learned.

-
- Q7** **A T cell meets its antigen on a resting tissue cell. What happens, and why is that useful?**
- It receives signal 1 without signal 2 and becomes anergic, permanently unresponsive to that antigen. Resting cells carry no B7, so a self-reactive T cell is silenced by meeting its self-antigen. The two-signal rule is peripheral tolerance.
-
- Q12** **Why can class II hold a peptide of 20 residues when class I cannot?**
- The class I cleft is closed at both ends and grips the peptide's termini, so only 8-10 residues fit. The class II cleft is open and grips the middle. The proteasome cuts to a standard size; cathepsins do not.
-
- Q28** **Why does IL-1 β need the inflammasome when TNF does not?**
- IL-1 β has no signal peptide and is made as an inactive 31 kDa pro-form. Signal 1 writes the pro-form and stops. Signal 2 assembles the inflammasome, caspase-1 cuts it to 17 kDa and opens the GSDMD pore. TNF has a signal peptide and needs none of this.

Three sentences carry both lectures.

- 1 A CD molecule is **half of a pair**. Name the partner and the cell it sits on, and you have named the fact.
- 2 An MHC molecule is **a display case**. Ask where the peptide came from, and the class, co-receptor and T cell follow.
- 3 A cytokine is **an instruction**, and no cell obeys it until a second, independent signal confirms the danger is real.